

Individual Progress Report

Group Topic: **Urban Air Quality & Environmental Health**

Lab No. Week 3

Git Repo/ Colab Link:

Date: 05/08/26

<https://github.com/Cristy16/BDA---Final-Project>

I. Objectives

General Objective

To establish the technical foundation for the machine learning pipeline, justify the analytical framework, and evaluate the **Random Forest Regressor's** performance on the validated Urban Air Quality dataset.

Individual Objective (Setup & Justification)

- **Infrastructure & Data Mapping:** Initialize a dedicated Jupyter Notebook to ingest Phase 5 data, defining the feature matrix (X) and target vector (y) for PM2.5 estimation.
- **Methodological Rationale:** Provide a data-driven justification for selecting **Prediction (Regression)** as the optimal tool over alternative methods like Clustering or Classification.
- **Performance Validation:** Quantify the predictive accuracy of the tuned model using a 134,972-row test set and communicate findings through interpretive visualizations.

II. Tasks Completed

- **Notebook Architecture:** Authored a standalone Jupyter Notebook, transitioning from data loading (Phase 5 CSV) to the definition of input features (CO, NO₂, SO₂, and location) and the target variable (PM2.5).
- **Strategic Justification:** Documented the rationale for choosing a **Prediction** framework. While Clustering identifies patterns and Classification categorizes levels (e.g., "Good" vs "Bad"), I proved that Regression is the most valuable tool here because it captures the precise numerical relationships between gases and pollution concentrations.
- **Metric Computation:** Validated the fine-tuned Random Forest Regressor (parameters: n_estimators=50, min_samples_split=10). The model's performance was calculated using R², RMSE, and MAE against the withheld test set.
- **Visualization & Documentation:** Produced a *Predicted vs. Actual* scatter plot and a *Feature Importance* chart. Integrated these into the notebook alongside documentation on data ethics and metric definitions.

III. Task to be completed

I have successfully completed the setup, justification, and evaluation components of this phase and will now handover the finalized notebook for integration into our team’s Git repository.

IV. Results and Analysis

Model Performance Summary

Metric	Value	Interpretation
R^2 (R-Squared)	0.5237	Explains ~52% of PM2.5 variance.
RMSE	7.2692 \mu g/m^3	Average deviation from actual values.
MAE	5.3445 \mu g/m^3	Mean absolute error across the test set.

Analytical Insights

- **Framework Efficacy:** The decision to use **Regression** was validated by the model's ability to interpret complex gas correlations. Specifically, the high correlation between CO and NO_2 (0.78) identified in earlier phases allowed the Regressor to estimate PM2.5 with a precision that Classification would have oversimplified.
- **Predictive Drivers:** Feature importance analysis confirms that **Carbon Monoxide (CO)** is the dominant predictor. This alignment with our exploratory data analysis (EDA) suggests the model is correctly learning physical chemical relationships rather than noise.
- **Error Distribution:** The scatter plot shows high accuracy in the low-to-moderate range, though variance increases during extreme pollution spikes. This limitation is documented as being tied to external factors (weather/traffic) not present in the current dataset.

V. Challenges and Solutions

- **Challenge:** Encountered a *NameError* for *y_test* and other upstream variables when attempting to run the evaluation logic in isolation.
- **Solution:** Established a sequential execution protocol within the notebook ("Run All Cells"). This ensured the environment was fully initialized with the correct data splits and fine-tuned parameters before triggering the evaluation suite.

VI. Conclusion

The transition to Phase 6 was successful. By properly setting up the environment and providing a technical justification for **Regression**, the team has a clear understanding of why this model is the most effective choice for air quality forecasting. The final evaluation confirms a significant performance lift over the baseline, and the notebook is now prepared for final integration into the team repository.